

Transfer of spinal cord material to subsequent bovine carcasses at splitting.

During the slaughter process, cattle carcasses are split by sawing centrally down the vertebral column, resulting in contamination of each half with spinal cord material. Using a novel method based on a real-time PCR assay, we measured saw-mediated tissue transfer among carcasses. Up to 2.5% of the tissue recovered from each of the five subsequent carcasses by swabbing the split vertebral face came from the first carcass to be split; approximately 9 mg was spinal cord tissue. Under controlled conditions in an experimental abattoir, between 23 and 135 g of tissue accumulated in the saw after splitting five to eight carcasses. Of the total tissue recovered, between 10 and 15% originated from the first carcass, and between 7 and 61 mg was spinal cord tissue from the first carcass. At commercial plants in the United Kingdom, between 6 and 101 g of tissue was recovered from the saw, depending on the particular saw-washing procedure and number of carcasses processed. Therefore, if a carcass infected with bovine spongiform encephalopathy were to enter the slaughter line, the main risk of subsequent carcass contamination would come from the tissue debris that accumulates in the splitting saw. This work highlights the importance of effective saw cleaning and indicates that design modifications are required to minimize the accumulation of spinal cord tissue debris and, hence, the risk of cross-contamination of carcasses.